



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: COMPUTER APPLICATIONS [SET A]

FULL MARKS: 100
TIME: 2 HOURS

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

*This paper consists of **six** printed pages. This Paper is divided into two Sections.*

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

SECTION A(40 Marks)

(Attempt all questions from this Section)

Question 1

[20x1=20]

Choose the correct answers to the questions from the given options

(i)



Identify the feature of Java depicted by the above picture:

(a) Encapsulation (b) Data Abstraction (c) Inheritance (d) Class And Object

(ii) `___ a="1223422".indexOf('2',2);`

Select the data type and the value stored in a:

(a) String and 5 (b) int and 5 (c) String and 2 (d) int and 2

(iii) Choose the correct output:

`System.out.println(Math.pow((Math.max(Math.ceil(-9.8),Math.floor(-9.5))),2));`

(a) 81.0 (b) 81 (c) 72.0 (d) 64

(iv) The array `float abc[ ]` of size **5** will occupy:

(a) 320bits (b) 40bits (c) 160bits (d) 80bits

(v) Choose the correct output:

`System.out.println(String.valueOf(2589).charAt(2));`

(a) Error (b) Null (c) 5 (d) 8

(vi) Method prototype that **accepts two integer and returns an integer array**:

(a) `int fun(int a, int b)` (c) `int [ ] fun(int a,b)`
(b) `int [ ] fun(int a, int b)` (d) `int arr [ ] fun(int a, int b)`

- (vii) Choose the correct output:
System.out.println(Integer.parseInt("134")+ 'a');
 (a) 231 (b) 134a (c) 199 (d) "134a"
- (viii) Choose the correct output if **n=4**:

```
static void fun1(int n)    {
    int i, f;
    for(i=1,f=1;i<=n;i++,f*=i);
    System.out.print(f);    }
```

 (a) 120 (b) Error (c) 1 (d) 24
- (ix) Choose the correct output:
String a1="Success", b1="Happiness";
System.out.println(a1.substring(4).equals(b1.substring(6)));
 (a) ess (b) ness (c) true (d) false
- (x) **Assertion(A):** An infinite loop is an instruction sequence that loops endlessly.
Reason(R): In the infinite loop the test condition is always true.
 (a) Both **Assertion (A)** and **Reason (R)** are true and (R) is a correct explanation of (A)
 (b) Both **Assertion (A)** and **Reason (R)** are true and (R) is not a correct explanation of (A)
 (c) **Assertion (A)** is true and **Reason (R)** is false
 (d) **Assertion (A)** is false and **Reason(R)** is true
- (xi) Which of the following is **not a token**:
 (a) operator (b) byte code (c) literals (d) identifiers
- (xii) An array is passed by _____ to a user defined function.
 (a) value (b) identifiers (c) reference (d) index
- (xiii) **Assertion(A):** Method should be called explicitly either with object reference or class reference.
Reason(R): Method can be void or may have return types.
 (a) Both **Assertion (A)** and **Reason (R)** are true and (R) is a correct explanation of (A)
 (b) Both **Assertion (A)** and **Reason (R)** are true and (R) is not a correct explanation of (A)
 (c) **Assertion (A)** is true and **Reason (R)** is false
 (d) **Assertion (A)** is false and **Reason(R)** is true
- (xiv) Technique of **binding data and methods together** to keep safe from unauthorized access
 (a) Encapsulation (b) Data Abstraction (c) Inheritance (d) Polymorphism
- (xv) A **set of similar classes** is called a:
 (a) Encapsulation (b) Package (c) Inheritance (d) Polymorphism
- (xvi) A method that needs to be **called with the help of an object** :
 (a) void (b) return (c) non-static (d) static

- (xvii) A **constructor generated by the compiler itself** is called _____ constructor.
 (a) parameterized (b) default (c) non-parameterized (d) copy
- (xviii) _____ is an **instance of a class**.
 (a) method (b) class (c) main() (d) object
- (xix) The value returned by **Float.parseFloat("-321")**;
 (a) 321.0 (b) -321.0 (c) 321 (d) "321"
- (xx) A method that reverses a given number passed as an **Integer object as a method argument**:
 (a) Pure method (b) Impure method (c) Constructor (d) Overloading

Question 2

- i. Write the **Java expression** for $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ [2]
- ii. **Evaluate** the expression: $p += ++p + --p / ++p + p$; where $p=5$; [2]
- iii. Give the **output** of the following code-snippet: [2]

```
int arr[][]={{1,2,3},{4,5,6},{7,8,9}};
System.out.println(arr.length);
System.out.println(Math.pow(arr[2][2], arr[0][2]));
```
- iv. Give the **output** of the following code-snippet: [4]

```
String str = "Computer Applications";
String str1 = "Computer";
if (str.startsWith("C"))
    System.out.println("Start Same");
if (str.endsWith("T"))
    System.out.println("End Same");
System.out.println(str1.compareTo(str));
String str2 = str.substring(12).replace('i','@');
System.out.println(str2);
```
- v. Find the **error and rewrite** the code [2]

```
static add() {
    int a,b;
    a=2;
    a+=b;
    System.out.print(a+ " "+b); }
```
- vi. **Rewrite** the following code using the **do-while loop**: [2]

```
int s,a;
for(s=1,a=12;a>=3;a -=2)
    s*=a;
System.out.println("Result="+s);
```
- vii. **Fill in the numbered spaces** with correct statements: [2]
 A student wants to sort an array using bubble sort, but while writing the method few statements in the code goes missing. Help to run the bubble sort code successfully filling

the missing terms in it.

[void bubbleSort(int arr[]) is a method that accepts an array and sorts the array in ascending order]

```
void bubbleSort(int arr[])    {
    int n = ?1?
    for (int i = 0; i < n - 1; i++)
    for (int j = 0; j < ?2?; j++)
        if (?3?)                {
            int temp = arr[j];
            arr[j] = ?4?
            arr[j + 1] = temp;    } }
```

- viii. With the help of an example show the **scope of a local variable and a class variable** in Java program. [2]
- ix. Give differences between **break** and **System.exit (0)**. [2]

SECTION B (60 Marks)

(Answer any four questions from this Section)

The answers in this section should consist of the programs in either BlueJ environment or any program environment with java as the base. **Each program should be written using variable description/mnemonic codes** so that the logic of the program is clearly depicted. **Flowcharts and algorithms are not required.**

Question 3

[15]

Riverside transport company charges for the parcels of its customers as per the following specifications given below:

Class Name : **Riverside**

Member variables :

String name : **to store the name of the customer**

double w : **to store the weight of the parcel in kg**

double charge : **to store the charge of the parcel**

Member functions :

void accept() : to accept the name of the customer, weight of the parcel from the user

void calculate() :	<table><tr><th>Weight in kg</th><th>Charge per kg</th></tr><tr><td>Upto 10kgs</td><td>₹ 25 per kg</td></tr><tr><td>Next 20kgs</td><td>₹ 20 per kg</td></tr><tr><td>Above 30kgs</td><td>₹ 10 per kg</td></tr></table>	Weight in kg	Charge per kg	Upto 10kgs	₹ 25 per kg	Next 20kgs	₹ 20 per kg	Above 30kgs	₹ 10 per kg
Weight in kg	Charge per kg								
Upto 10kgs	₹ 25 per kg								
Next 20kgs	₹ 20 per kg								
Above 30kgs	₹ 10 per kg								

A **surcharge of 5%** is charged on the bill.

void print() : to print the name of the customer ,weight of the parcel, total bill inclusive of surcharge in a tabular form in the following format:

Name	Weight	Bill amount
-----	-----	-----

Define a class with the above mentioned specifications, create the main method, create an object and invoke the member methods.

Question 4

[15]

Define a class to accept values into an **integer array of order 4×4** and check whether the **sum of boundary element is equal to product of its non-boundary element** or not. **Display appropriate message.**

Example:

1	2	3	4
12	1	13	5
11	2	3	6
10	9	8	7

Sum of the boundary elements = $1+2+3+4+12+5+11+6+10+9+8+7=78$

Product of the non-boundary elements = $1 \times 13 \times 2 \times 3=78$

Question 5

[15]

Define a class to **accept a day of a week in number** and **a time in second's**. Display **the day in words** and **convert seconds into Hours, Minute and Seconds** and print the results in following format.

Input:

Day: 3

Seconds:12035

Output:

Wednesday

3Hours 20Minutes 35 Seconds

Question 6

[15]

Riddhi recently came to know about the **ascii values** but she is **not aware of the ascii values of the Alphabet series**. She wants to play a game that will help her to understand **the relation between ascii values and alphabets**. So design a **class Letters** and **accept letters (in Upper Case) at random** from Riddhi and **store in a single dimensional array of size 10**. **Accept any ascii value from her**, where the ascii will be **≥ 65 and ≤ 90** and **display the character that matches the ascii value**. Also **display the frequency of the letter in the given array**. **Display “Try again” if the letter is not present in the array**.

Choose an appropriate Searching technique for the above program.

Examples:

Input:

A	Q	K	J	R	Q	L	M	B	D
---	---	---	---	---	---	---	---	---	---

Ascii= 81

Output:

Letter	Frequency
Q	2

[15]

Question 7

Design a **class to overload** a function `fun( )` as follows:

void fun(): Method to display the following pattern

```
1
2 4
3 5 7
6 8 10 12
9 11 13 15 17
```

void fun(int num): to accept a number and check whether it is a Harshad number or not. (A harshad number , is an integer that is divisible by the sum of its digits.)

Example:

Number 200 is a Harshad Number because the sum of digits 2 and 0 and 0 is 2(2+0+0) and 200 is divisible by 2. Number 171 is a Harshad Number because the sum of digits 1 and 7 and 1 is 9(1+7+1) and 171 is divisible by 9.

[Do not write the main method.]

Question 8

Write a program that **accepts a sentence as input. Modify each word of the sentence** by bringing **all vowels of the word at the beginning followed by the constants.**

Display the **new sentence with the modified words.**

Example:

Input: Understanding Computer Applications

Output: Ueaindrstndng oueCmptr Aiaiopplctns